

The City College of New York
EE 22100 Electrical Engineering Lab 1
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Lab Report Experiment 7: Digital Clock

I. Aim/Objective:

The objective of this experiment was to design and build a functional 12-hour digital clock using Binary Coded Decimal counters, which are a type of integrated circuit. The purpose was to understand how these BCD counters interacted with the seven-segment displays, specifically how the BCD data is used to direct the displays and show the correct time, allowing us to display the seconds, minutes, and hours accurately. The experiment also aimed to show us how AND gates function, how integrated circuits are used in these digital designs, and how to create a fully functional digital clock that counts seconds, minutes, and hours.

II. Introduction:

In this experiment, we built a functional 12-hour digital clock using **binary-coded decimal counters**, which are a type of **integrated circuit**. These ICs are specifically designed to handle BCD numbers, where each decimal digit is represented by a 4-bit binary number. The **CD4510** is a pre-settable BCD counter, allowing it to be set to specific values, while the **CD4518** is a ripple counter that automatically increments when clock pulses are received. These counters were important and allowed for the counting of the seconds, minutes, and hours of the clock. The BCD counters convert binary data into a decimal format, which is then displayed on the **seven-segment displays** connected to the counters. The seven-segment displays light up in specific patterns according to the BCD values, displaying the numbers on the clock. When the BCD counter reaches a specific value, the display shows the matching digit. The BCD counters count from **0 to 9**. By converting the binary count to decimal, the clock's time is displayed accurately on the display screen. AND gates were also necessary to complete this digital clock. An **AND gate** is a logic gate that was used to cascade the counters and count increments, making sure that when one counter, being seconds or minutes, reached its maximum value of 60 (only displaying 59 on the screen), it triggered the next counter, minutes or hours, to increment, keeping the time accurate and synchronized in the 12-hour format. For example, when the seconds counter reached a full cycle, the AND gate made sure that the minute counter increased by 1. Similarly, when the minute counter reached 60, the AND gate made sure that the hour counter increased by 1. The AND gates managed the transition from one counter to the next, making sure that the clock properly tracked the numbers one after the other. The **cascading effect** of the counters, produced by the AND gates, is important for the clock to function. Cascading means that when one counter reaches its maximum value, it triggers the next counter to increment. Using **integrated circuits (ICs)** like the BCD counters and AND gates made the digital clock design simpler and more reliable. These ICs come pre-made with multiple logic gates and components already built into them, which makes it easier to set them up on the breadboard without having to wire everything without the ICs. This experiment showed how these components work together to create a functional digital clock. By learning how the BCD counters and AND gates work, we were able to build a fully operational 12-hour clock.

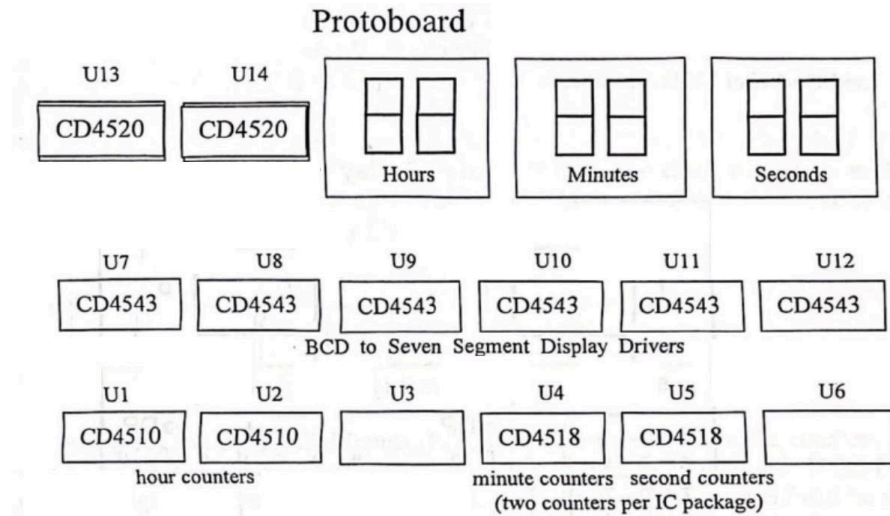


Figure 1: Digital Clock Circuit Layout on Protoboard

- This figure shows the layout of a digital clock circuit, with counters for seconds, minutes, hours, and Integrated Circuits that are meant to control the seven-segment displays.

III. Components

- **Seven-segment displays:** For time display

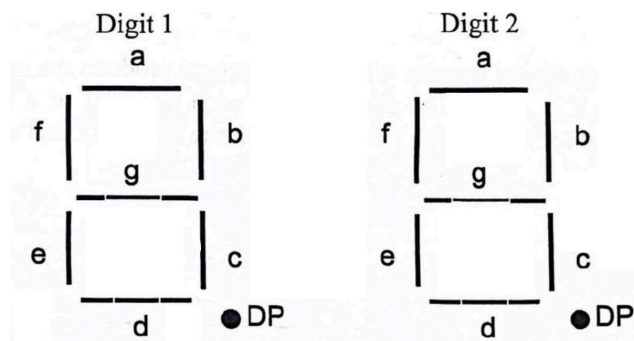


Figure 2: Segment Labeling of Two-Digit Seven-Segment Display

- This figure shows how the segments of two seven-segment displays are labeled (a to g and DP). Each display shows one digit (Digit 1 or Digit 2) by lighting up specific segments

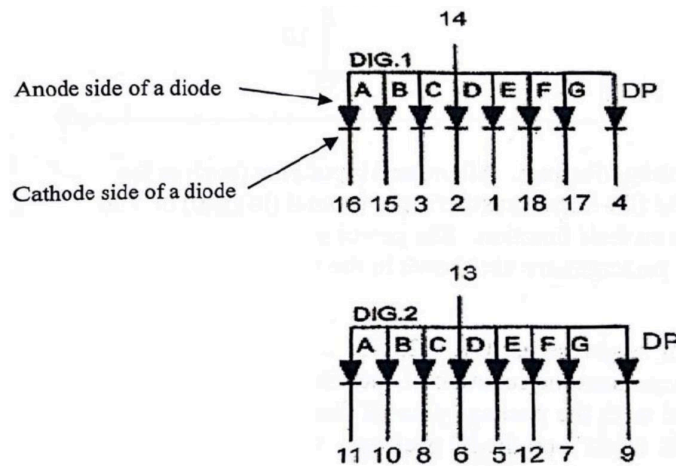


Figure 3: Circuit Diagram for Two-Digit Seven-Segment Display

- This figure shows how the segments of the two displays are connected, with shared anodes and individual cathodes to control each segment.
- **CD4543:** BCD-to-seven-segment display Integrated Circuit

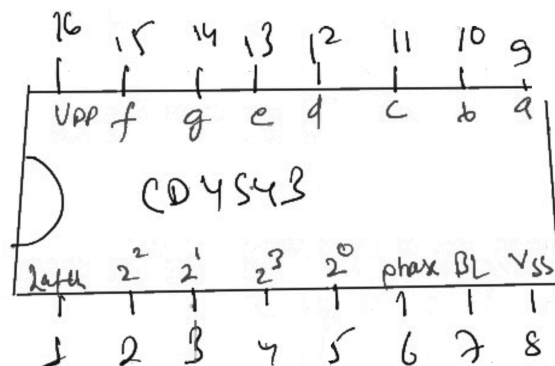


Figure 4: Pinout Diagram for CD4543 to Seven-Segment Display

- This figure shows the pin configuration for the CD4543 integrated circuit, including input bits (2⁰ to 2³), segment outputs (a to g), and control pins for phase, blanking (BL), and power connections (VDD and VSS).

- **CD4518:** Dual BCD counter

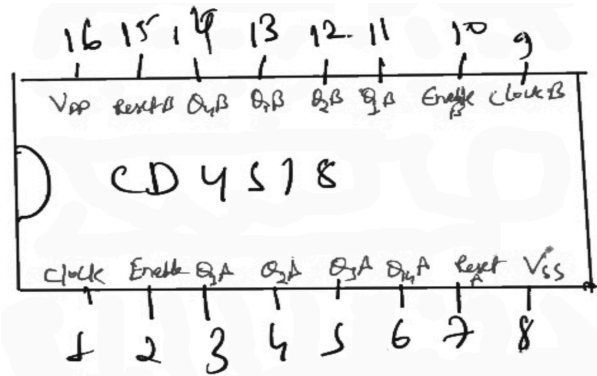


Figure 5: Pinout Diagram for CD4518 Dual BCD Counter to CD4543

- This figure shows the pin layout of the CD4518 dual BCD counter, with inputs for clock, enable, and reset, and outputs (QA to QD) for two counters (A and B). It also includes power connections (VDD and VSS).

- **CD4510:** BCD pre-settable counter

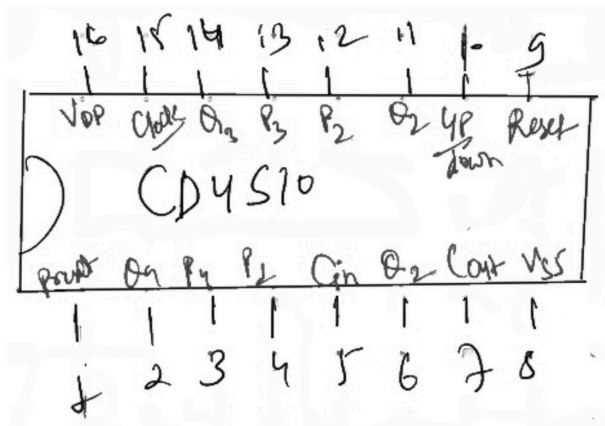


Figure 6: Pinout Diagram for CD4510 BCD Pre-Settable Counter

- This figure shows the pin layout of the CD4510, a BCD pre-settable counter with clock, reset, and up/down control inputs, pre-settable inputs (P1 to P4), and outputs (Q1 to Q4) for counting operations.

- **CD4081:** AND gate Integrated Circuit

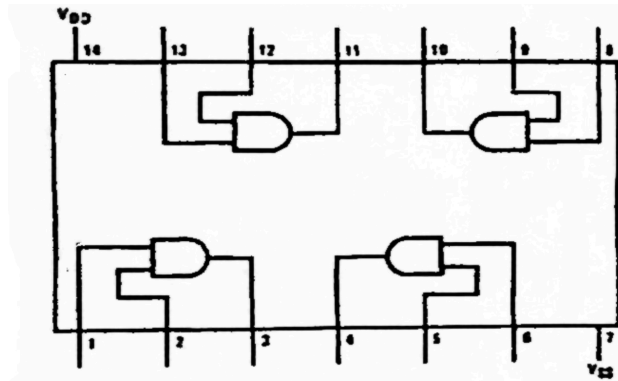


Figure 7: Diagram of CD4081 AND Gate Integrated Circuit

- This figure shows the CD4081 AND gate IC, which contains four independent 2-input AND gates. The first pin (VDD) is the positive power supply, and the last pin (VSS) is ground. Each gate has two inputs and one output.

- **Function generator:** For clock signal input

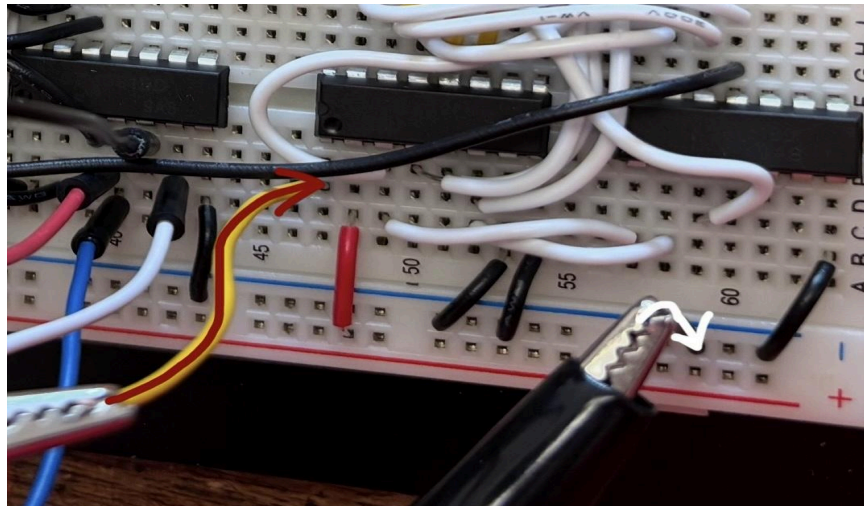


Figure 8: Function Generator Connection

- This figure shows the function generator connected to the bottom pin (pin 1) of the seconds counter (CD4518) for the clock signal and the ground.

We used a function generator to provide a stable frequency signal that served as the clock pulse for our circuits. The frequency of the function generator was set to 1 Hz, allowing the counters to increment at a realistic rate for a digital clock. The wave type was set to a square wave. The voltage levels were configured with a low of 0 volts and a high of 5 volts to match the Vdd of the design. The output of the function generator was connected to pin 1 (the clock input) of the seconds IC circuit CD4518 and also to the ground. After setting up the seconds section, we needed another function generator for the minutes IC to cascade it. For the minutes, the output of the AND gate from the seconds section was connected to pin 1 (the clock input) of the minutes IC circuit (CD4518), allowing it to cascade. Without the function generator, the counters would not receive the clock pulses needed to increment, and the digital clock would fail to count the seconds, minutes and hours.

- **Power supply:** 5V battery

Decimal Equivalent	0	1	2	3	4	5	6	7	8	9	x	x	x	x	x	x
BDC	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	x	x	x	x	x	x

Figure 8: BCD Code Table

- This figure explains how decimal digits (0–9) are represented using four binary bits in binary-coded decimal (BCD). It also describes how BCD counters like CD4510 and CD4518 are used to count and drive seven-segment displays for digital clocks and other applications.

IV. Challenges and Troubleshooting

Week 1: We began by working on the seconds portion of the clock, setting up the first display and connecting two CD4543 integrated circuits beneath it. We also connected a CD4518 binary-coded decimal counter to handle the counting. We then added two 4081 AND gates to combine the inputs and ensure the counter would increment by 1 with each clock pulse. Initially, we used one AND gate but later decided to add another due to issues with the first setup. After connecting everything, the seconds counter only counted up to 9, and despite checking the wires, we couldn't figure out the issue. After redoing the circuit and checking the connections carefully, the counter stopped working entirely. Instead of counting properly, the display would jump randomly between numbers. We ran out of time, so we planned to attempt the fix in Week 2.

Week 2: In Week 2, we continued troubleshooting the seconds section, but despite our efforts, it wasn't working as expected. We carefully checked every connection and even swapped out ICs, suspecting a faulty component. We also considered that the breadboard might have a defect. After testing these possibilities, we realized that we had forgotten to connect the power and ground wires to the AND gates. This simple mistake was preventing the gates from working correctly. Once we fixed that, the seconds counter worked perfectly. With the seconds section functioning, we moved on to the minutes section. We carefully connected the seconds counter to the minutes counter and checked the wiring. However, we ran out of time before we could finish testing the minutes section.

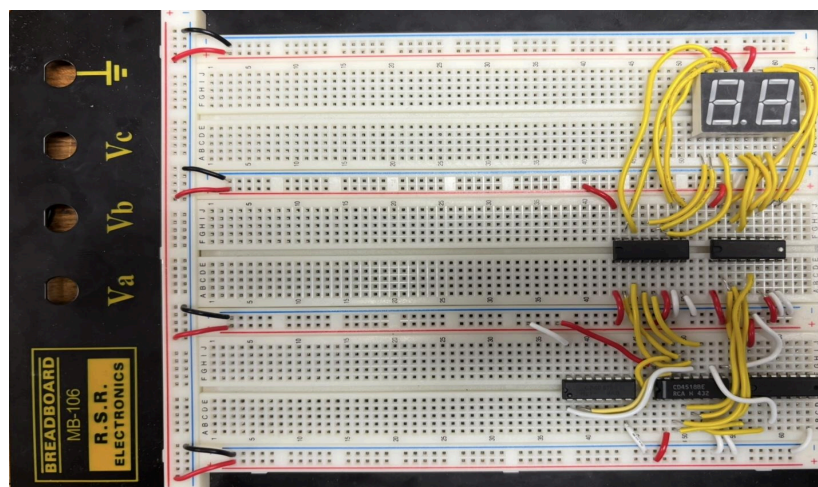


Figure 8: Breadboard setup for Seconds

- In this setup, two **CD4543** ICs were connected to the seven-segment display, while the **CD4518** IC was used for the seconds counter, placed underneath the **CD4543**. To control the counting, two **4081 AND gates** were placed to the left and right of the **CD4518**. This breadboard setup worked as expected, allowing the seconds counter to function correctly and the display to show the time accurately.

Week 3: We completed the wiring for the minutes section and tested it. Initially, it seemed to be working because the minutes were counting up, but we quickly noticed that the minutes would only count up to 10 instead of 59. The seconds were working fine, but the minutes counter was stopping at 10. After carefully checking the connections, we discovered that the AND gate inputs for the seconds-to-minutes connection were reversed meaning that each corner of an AND gate takes a different position for the input and output. Now learning how to use multiple inputs for the AND gate we fixed this and the minutes counter worked as expected, counting up to 59. We then began working on the hours section but we didn't finish it by the end of the lab day.

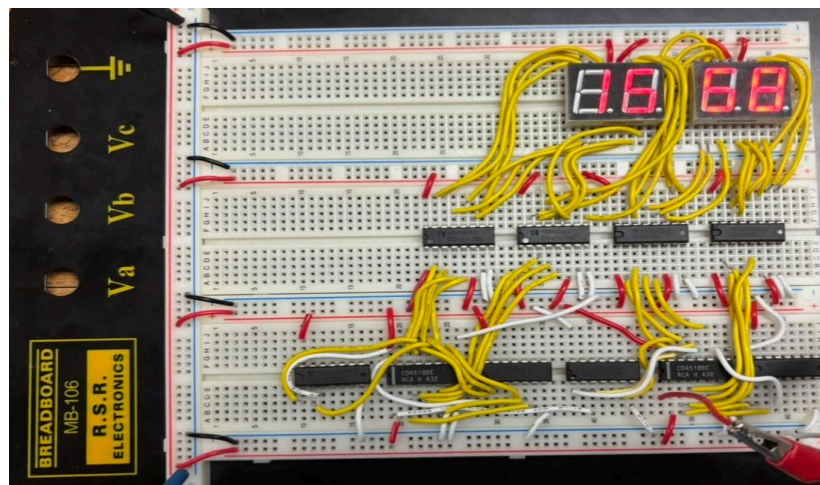


Figure 9: Breadboard setup for minutes

- In this setup, two **CD4543** ICs were connected to the seven-segment display, and the **CD4518** IC was used for the minutes counter, placed beneath the **CD4543**. Two **4081 AND gates** were positioned to the left and right of the **CD4518** to control the counting process. The function generator of the minutes **CD4518**'s clock pin (pin 1), was connected to one of the outputs of the seconds **AND gate** to enable cascading. This setup worked as expected, allowing the minutes counter to function correctly, with the display showing the time accurately.

Week 4: When we continued working on the hours section, everything was plugged in correctly, but the display wouldn't turn on. While supplying the breadboard with power, the voltage from the power supply dropped from 5V to 0.2V when plugged into the + and -, indicating an issue. After carefully inspecting the setup, I realized some wires had started to rip, which was preventing the circuit from functioning properly. The tightness of the wires caused them to stretch and become damaged, leading to disconnected connections. I quickly realized I needed to restart the entire setup to address the wiring issue, as I couldn't pinpoint which wire was causing the problem and the breadboard was not receiving the necessary 5V. After removing everything and starting again with new wires for the seconds section, the breadboard was receiving the correct 5V, and the seconds counter worked as expected. I then rebuilt the minutes section, which also functioned properly. However, when I moved on to the hours section, the display turned on but showed a line in the middle of the "0" displaying an "8" instead. At first, I thought it might be a wiring issue, so I double-checked all the connections. When that didn't resolve the problem, I began to believe the breadboard itself might be the issue. I swapped the display with a new one, which removed the dash in the "0," but the new display flickered on and off and wouldn't stay lit. At this point, I believed the issue was with the specific row on the breadboard. To further troubleshoot, I moved everything over, shifting the components down by one position, and rebuilt the hours section. This time, the hours section turned on and began counting properly, but it worked only if I kept my hand pressed down on the wires to prevent them from coming loose. If I held the wires down, it would continue to function correctly.

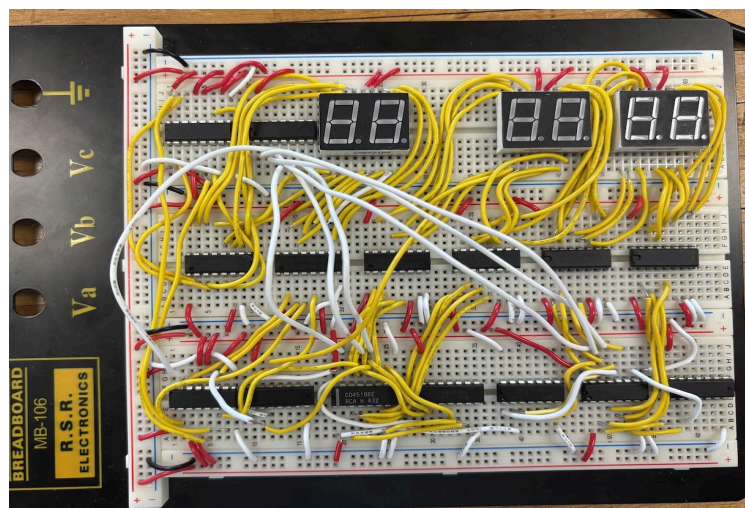


Figure 10: Breadboard for Hours (Non-Working Setup)

- In this setup, two **CD4543** ICs were placed under the display, with two **CD4510** ICs positioned beneath the **CD4543** for the hours counter. One **4081 AND gate** was placed to the left, and another near the display. The left **CD4510** IC was connected to the first AND gate, and the right **CD4510** IC was connected to the second AND gate. To cascade the counters, the output from the minutes AND gate was connected to the clock input of the right **CD4510** IC for the hours. Additionally, both the left and right **CD4510** ICs were connected to the input of one AND gate, and the output was routed to the same output column. However, this setup did not work as expected, so I restarted the entire circuit due to damaged wires.

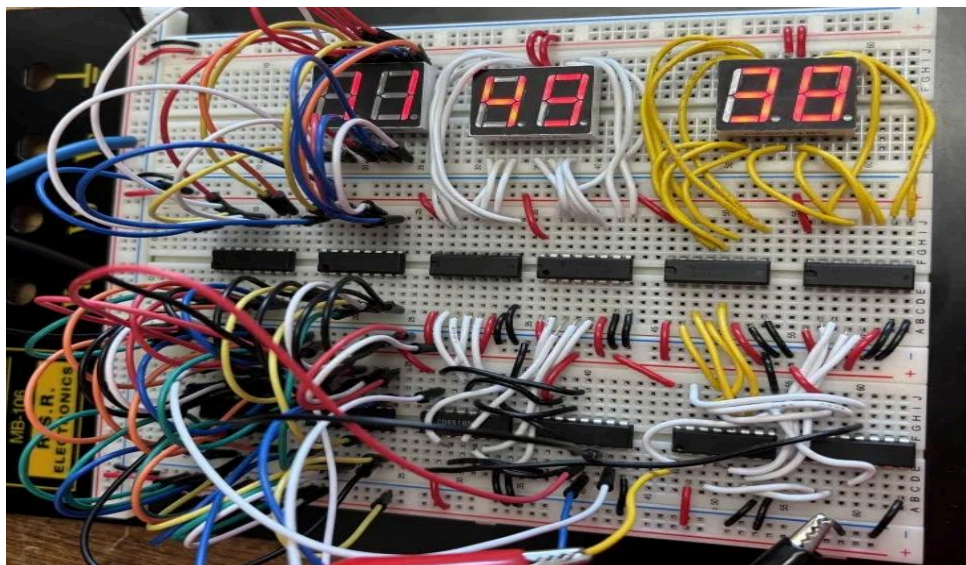


Figure 4: Breadboard for Hours (Working Setup)

- In the working setup for the hours section, the setup was similar to the non-working one, but with adjustments of using less gates and new wires. Two **CD4543** ICs were placed beneath the display, with two **CD4510** ICs below them. We used one **4081 AND gate** and connected the first input from the left **CD4510** and the other input from the right **CD4510**, and the output from both ICs was connected to the same AND gate output column. This setup allowed the hours section to function correctly and display the time accurately.

Successful Breadboard Attempts Videos:

Successful Attempt Seconds:

https://youtu.be/-XK_WN4RT6E

- This was our successful attempt at creating a seconds counter for our first attempted breadboard. In this video, you can see the seconds increase by one as they count from 0 to 59. Once they hit 59, they reset back to 0.

Successful Attempt Minutes:

<https://youtu.be/8R0py8RNqEQ>

- This video shows our successful attempt at creating a minutes counter that cascades perfectly with the seconds. You can see the minutes increase by one each time the seconds complete a cycle from 0 to 59. Once the minutes reach 59, they reset back to 0, demonstrating that we've successfully synced the minutes with the seconds.

Final Rebuilt Result Working Digital Clock:

https://youtu.be/8n2_f8gFhdw

- After our attempt of rebuilding everything worked out great, the seconds and minutes on the clock matched up, and our digital clock was working. This video is made up of several clips because, although we got most things fixed, the wires were still a bit loose. Every once in a while, we had to hold them down during the recording to keep everything connected.

Report Questions:

1. An explanation of how you made the counters count to the correct numbers.

- In this design, the counters were created using an integrated circuit designed for counting binary counters. These counters automatically increment the count with each clock pulse. The counting logic uses AND gates to manage the timing and control when the count should increment. The counter ICs were connected to the clock pulse generator, so each pulse triggered the counter to increment, which led to the correct counting behavior.

2. You will describe the display and how it works.

- The display used in the clock is a 7-segment LED display. The output from the counters was decoded using the AND gates to determine which segments to light up for each digit. As the counters incremented, their values were changed into the appropriate segment patterns to display the correct number on the LED display.

3. Describe the base time generator

- A base time generator provides a consistent clock signal or a pulse that guides the rest of the circuit. The base time generator determines the time interval between each pulse that increments the counter.

4. Tell your reader how long a 9V battery or 6 AA batteries will run your clock.

Discuss your findings about how much current the logic portion of the clock draws compared to the displays. Do you know of another type of display that draws much less current than the display that you used?

- A 9V battery or six AA batteries will probably run the clock for a few hours. The 7-segment LED display uses a significant amount of current, especially when multiple of the segments are lit. A more reliable alternative to the 7-segment LED display could be an OLED display, since it consumes way less power than LED displays, making this a better choice when having a limited power source.

V. Conclusion:

This lab taught me how to build a 12-hour digital clock using BCD counters and seven-segment displays. I learned how to connect components like the CD4518, CD4510, and CD4543 to create a working clock that tracks seconds, minutes, and hours. I also learned how to reset and cascade the counters to ensure the clock works in a continuous loop, updating each time unit as the clock progresses. During the lab, I faced challenges with wiring, faulty components, and syncing the counters. However, by carefully troubleshooting and adjusting the setup, I was able to fix these issues and get the clock working properly. This lab helped me understand how digital circuits work and showed me the importance of testing and making corrections when things went wrong. These skills will be useful in future projects.